

EQ

EQ Part I - Why EQ?

- Balance frequency spectrum
- Shape presentation (sharp, rounded, dirty, clean)
- Separation plus its subset definition- no separation=no def
- Mood- bright,dark.
- FX
- Depth=less hf=distancing. EQ and depth-lpf swept up/down
- Fine level adjustments. Boost or attenuate part of spectrum.

EQ Part 2- Freq Spectrum

- Humans capable of hearing 20Hz to 20 kHz
- Freq perception not linear
- We do not perceive 100 Hz as we do 300Hz
- Sections of freq referred to as bandwidth
- 110Hz is an 8ve below 220Hz. 220Hz is an 8ve below 440hz. As freq gets higher the space between freq gets less.

EQ Part 3- Low bands

- Subsonic=up to 20Hz=feel not hear
- Low bass=20-60Hz=lows=kick and bass have fundamentals here=still feel not hear=ex. Lowest piano.
- Mid bass=60-120Hz=start to perceive pitch/tonality= power frequencies=depth
- Upper bass=120-250Hz=most instruments have fundamentals in this range=depth.Alter natural tone here.

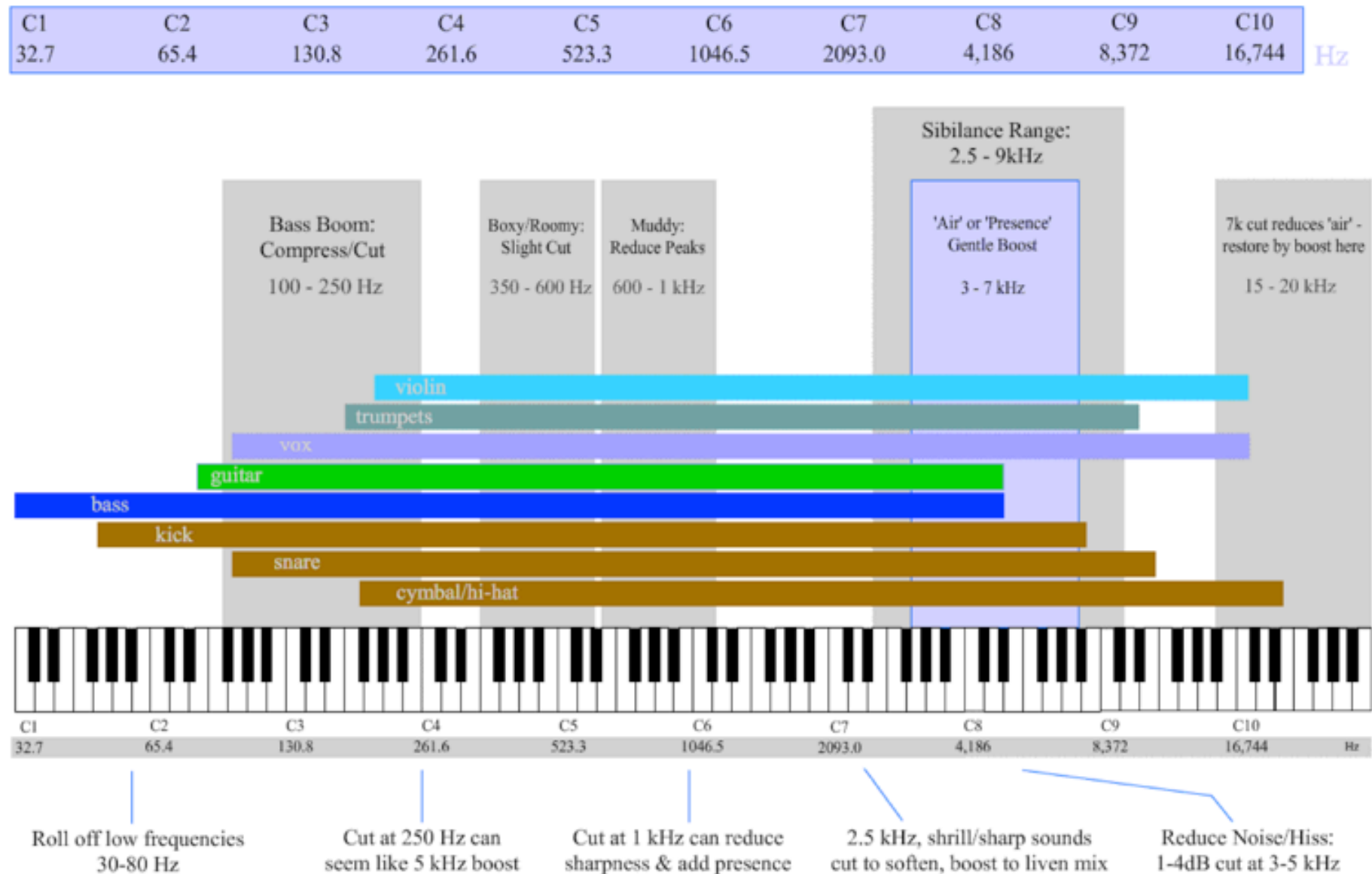
EQ Part 4- Mid/High bands

- Low-mids=250Hz-2kHz=low order harmonics of many sounds=meat, color and timbre=some depth.
- High-mids=2-6kHz=human's most sensitive range=complex harmonics=loudness, definition, presence, dialogue.
- Highs=6-20kHz=not much energy=air, brilliance, effervescence.

EQ Part 5- Freq Chart

Podcomplex Frequency Overview Chart

A graphical reference sheet for arrangement, equalisation and mastering



EQ Tips #1

- Always EQ in context
- A little bit goes a long way unless using eq for effects
- The more gain you apply in EQ the more you increase the phasing artifacts.
- Try to use EQ to attenuate instead of EQ to boost
- Use pass filters to remove content from extremes while parametric filters are more surgical

EQ Tips #2

- HPF's increase clarity and definition=help to stand out in the mix. Removes dispensible lows that clear room for other low frequencies. Our brains easily reconstruct missing fundamentals.
- HPF's can reduce size and depth of reverbs
- LPF's remove hiss or HF noise
- LPF's mark HF boundary of HF sounds
- Use several passes of eq as opposed to one for more control.

Listening

- **Eq and depth**

Listening

- **Eq and depth**

Listening

- **Drums source**

Listening

- **Drums source**

Listening

- HPF 50Hz

Listening

- HPF 50Hz

Listening

- HPF 100Hz

Listening

- HPF 100Hz

Listening

- LPF 6kHz

Listening

- LPF 6kHz

Listening

- LPF 250 Khz

Listening

- LPF 250 Khz

Listening

- **Vocal source**

Listening

- **Vocal source**

Listening

- HPF 50Hz

Listening

- HPF 50Hz

Listening

- HPF 100Hz

Listening

- HPF 100Hz

Listening

- LPF 6kHz

Listening

- LPF 6kHz

Listening

- LPF 250 Khz

Listening

- LPF 250 Khz